AI-Generated vs Human-Written Phishing Emails: A User Detection Study

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***Abstract*—This study evaluates the human ability to distinguish AI-generated phishing emails from legitimate ones. We conducted a survey and analyzed the results to compare human performance against automated systems.**

I. Introduction

A. Context and Problem

Phishing remains one of the most significant cybersecurity threats globally. With the advent of Large Language Models (LLMs), attackers can now generate highly convincing, error-free phishing emails that are difficult to distinguish from legitimate correspondence.

B. Gap Statement

While previous studies have focused on traditional, human-written phishing emails, there is a lack of empirical data on how well users can detect AI-generated phishing. It remains unclear if the typical "red flags" (grammar errors, awkward phrasing) are still effective.

C. Our Contribution

The main contributions of this study are:

- We conducted a survey with real participants to test their detection capabilities.
- We measured the confidence levels of participants when classifying emails.
- We compared human detection rates (Recall) with baseline automated systems.

II. Methodology

A. Experiment Design

Participants were presented with a mix of legitimate and AI-generated phishing emails. They were asked to classify each email. To ensure the reliability of the results, a warm-up phase was implemented, and the first few responses were discarded from the final analysis.

B. Metrics

To evaluate performance, we calculate the Accuracy and Recall of the detection. The Accuracy is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

where TP is True Positive, TN is True Negative, FP is False Positive, and FN is False Negative.

All data processing and statistical analysis were performed using Python, Pandas, and Seaborn.

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